

Joshua Plummer | Curriculum Vitae

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Professional Profile

Teaching-focused mathematician with experience in undergraduate mathematics instruction, curriculum development, and student success initiatives. Designs inclusive, active-learning environments supported by peer mentoring and co-requisite models to improve student outcomes. Recognized for instructional impact through student-nominated teaching honors.

Education

Doctor of Education | Instructional Systems and Learning Sciences

Auburn University at Montgomery, Montgomery, AL.

Admitted: April 2026

Master of Science | Mathematics

University of South Alabama, Mobile, AL.

Aug. 2016 - May 2018

Thesis: On Matroid Representability

Bachelor of Science | Mathematics and Statistics

University of South Alabama, Mobile, AL.

Aug. 2010 - May 2016

Bachelor of Science | Physics

University of South Alabama, Mobile, AL.

Aug. 2010 - May 2016

Experience

Auburn University at Montgomery

Lecturer

Montgomery, AL.

August 2024 - Present

- Teach undergraduate mathematics courses including Precalculus, Elementary Statistics, and College Algebra across in-person and asynchronous formats.
- Consistently received student evaluation ratings of 4.8/5.0 or above, among the highest in the Mathematics Department.
- Design and evaluate assignments, quizzes, and exams aligned with learning outcomes and equitable grading practices.
- Collaborate on departmental committees to standardize syllabi and assessments across multi-section courses (Precalculus, Elementary Statistics, College Algebra).
- Integrate instructional technology (e.g., LMS, GeoGebra, online tools) to support interactive, student-centered learning.
- Implement active learning and experiential approaches, contributing to improved student engagement and stronger performance as measured by DFW rates.

Purdue University Global

Adjunct Instructor

Remote

June 2025 - Present

- Teach fully online undergraduate mathematics courses.
- Adapt instructional materials for a diverse, nontraditional student population, many of whom are returning adults balancing work and family responsibilities.
- Facilitate learning through structured weekly modules, discussion-based engagement, and timely, individualized feedback.
- Support student progress via virtual office hours and targeted outreach using LMS analytics.

University of South Alabama

Adjunct Instructor

Mobile, AL.

May 2025 - Present

- Teach undergraduate mathematics courses in online and hybrid formats.
- Provide clear, actionable feedback on student work to support persistence and mastery in virtual environments.
- Leverage LMS analytics to identify at-risk students early and provide targeted intervention through personalized outreach.

St. Luke's Episcopal School

Upper School Teacher

Mobile, AL.

Aug. 2019 - Apr. 2023

- Developed and implemented engaging lesson plans using inquiry-based and active learning approaches.
- Applied the Experience First, Formalize Later model to promote conceptual understanding.
- Led curricular initiatives within the mathematics department, including the introduction of AP Calculus AB.
- Advanced the adoption of standards-based grading and active learning practices across courses.
- Sponsored Mu Alpha Theta (Math Honor Society).
- Coached the school esports team, fostering student engagement, teamwork, and strategic thinking.
- Taught 5–6 courses per semester ranging from Honors Chemistry to advanced mathematics.

University of South Alabama

Adjunct Instructor

Mobile, AL.

Aug. 2018 - Dec. 2022

- Taught 1–3 courses per semester ranging from Finite Mathematics through Calculus.
- Incorporated instructional technologies to support student learning and performance.
- Developed lesson plans adaptable to in-person, online, and blended modalities.

Faulkner University

Adjunct Instructor

Mobile, AL.

Jan. 2021 - May 2022

- Taught up to three courses per semester ranging from College Algebra to Precalculus.
- Assessed student work and provided timely feedback to support skill development and course success.

University of South Alabama

Graduate Teaching Assistant

Mobile, AL.

Aug. 2016 - May 2018

- Led recitation sessions, facilitated problem-solving discussions, and provided targeted examples.
- Created and graded weekly quizzes with feedback aligned to course objectives.
- Held weekly office hours and exam review sessions.

University of South Alabama

Lead Peer Academic Leader (PAL)

Mobile, AL.

Aug. 2012 - May 2016

- Mentored students and connected them with appropriate academic support resources.
- Supervised and evaluated a team of 5–6 PALs; supported onboarding and performance review.
- Interviewed and helped select prospective PALs.

Certifications and Training

NextGen Learning: AI Integration Series

Purdue University Global, 2026

- Developed strategies for integrating AI tools into course design to enhance student engagement and instructional efficiency.
- Applied tools including Google Gemini, NotebookLM, and Gamma AI for content creation, assignment design, and instructional support.
- Designed AI-informed learning activities that promote critical thinking and responsible AI use among students.
- Incorporated AI-supported approaches into instructional materials, including presentations, assessments, and interactive content.

Certificate Series in Accessibility Practices

Auburn University at Montgomery, 2026

- Completed a series of certifications in Microsoft accessibility practices.
- Training in accessible design for Word, PowerPoint, Excel, Canvas, and multimedia content.
- Emphasis on ADA-aligned practices including document structure, alt text, accessible tables, and meaningful link text.
- Utilized tools such as Panorama for accessibility remediation and compliance checking.

Enhancing Online Education through Instructor-Created Video Content

Purdue University Global, 2026

- Training in best practices for creating instructional video content to improve student engagement and retention.
- Developed skills in recording, editing, and delivering course content using Kaltura.
- Integrated video content within the Brightspace LMS to support online and asynchronous learning environments.
- Designed video-based assessments and completion criteria to monitor student progress and reinforce learning outcomes.

Flight School (Student Success & Peer Mentoring Training)

Auburn University at Montgomery, 2026

- Trained in implementing peer-assisted learning (co-pilot) and peer mentoring models to support student learning.
- Developed strategies to increase engagement, accountability, and course persistence.
- Emphasized inclusive teaching practices and structured support systems.

Certificate in Applied Data Analytics

Auburn University at Montgomery, 2025

- Applied Python for data wrangling, exploratory analysis, and visualization of real-world datasets.
- Implemented regression models, classification, and clustering techniques.
- Applied statistical and computational methods to real-world datasets to support data-driven decision-making.
- Tools: Python (pandas, numpy, matplotlib, seaborn, scikit-learn, statsmodels, scipy, plotly, Jupyter Notebook).

HHMI Grant-Supported Teaching & Student Success Training

Auburn University at Montgomery, 2024

- Training in evidence-based instructional strategies and inclusive pedagogy through an HHMI grant-supported initiative.
- Emphasis on active learning, mentoring, and student success practices.

Experience First, Formalize Later Workshop

Synchronous Online, 2022

- Training in an instructional approach that prioritizes conceptual exploration before formal mathematical abstraction.
- Emphasis on inquiry-based learning and developing student intuition prior to formalization.

FISH! Philosophy Workplace Training

St. Luke's Episcopal School, 2021

- Training focused on workplace culture, engagement, and positive communication strategies.
- Emphasis on attitude, accountability, and fostering collaborative environments.

AP Summer Institute: AP Calculus AB

Auburn University, 2020

- Completed College Board-endorsed training for teaching AP Calculus AB.
- Focus on curriculum design, alignment with AP standards, and preparation for the AP examination.

Leadership, Service, and Committee Work

2026 - Present: Social Committee | *Chair*

2026 - Present: Online Course Integrity Committee | *Member*

2024 - Present: Course Coordination Committees | *Member*

STAT 1070; MATH 1120/30/50; MATH 1050

2024 - Present: Co-requisite Support Committees | *Member*

STAT 1070; MATH 1120; MATH 1050

2025 - Present: Alabama State-Wide High School Mathematics Contest | *Member*

2025 - Present: SK-Day Committee | *Member*

2025 - 2026: Social Committee | *Co-Chair*

Awards and Honors

2026: Student-Nominated Significant Impact Certificate of Appreciation

2025: Student-Nominated Recognition for Instructional Impact

2024: Student-Nominated Significant Impact Certificate of Appreciation

2019: Conference of Southern Graduate Schools Outstanding Thesis Award

2016–2018: Graduate Assistantship

2017: Melton McLaurin Graduate Fellowship

2016: Recognition of Dedication and Leadership as a Lead JagPAL

2015: Chris Nash Mathematics Competition Scholarship

2012–2014: President of the Society of Physics Students

2010–2014: Freshman Physics Scholarship

2013: Induction into Sigma Pi Sigma

Projects and Publications

Co-requisite Workbook Development

Spring 2025

Contributed to the revision and redevelopment of corequisite workbooks for STAT 0070 (Elementary Statistics), MATH 0120 (Precalculus Algebra), and MATH 0050 (College Algebra) at AUM. Work included content review, refinement of explanations, and restructuring of practice activities to improve clarity, coherence, and alignment with learning objectives.

Graduate Thesis

Jan. 2017–May 2018

Wrote a Master's thesis titled, "On Matroid Representability" under Dr. Elena Pavelescu at the University of South Alabama. The thesis examined matroid representations, including Rota's conjecture, and presented novel proofs of selected results.

Academic Materials Development

May 2015 – Dec. 2015

Led a project under Mr. Ira Johnson while serving as a Lead JagPAL at the University of South Alabama to create educational materials for the Office of Student Success. Developed approximately 110 resources (e.g., quick-reference guides and example videos) for historically challenging general education courses including Calculus, Physics, and Chemistry.

Editorial on the Importance of General Education Requirements

Summer 2015

Written for the College of Arts and Sciences' freshman seminar course at the University of South Alabama. An edited version was published by the university newspaper — Plummer, J. (2016). *The Importance of General Education*. *The Vanguard*, 59(9), 7.

Astronomy Research

Aug. 2013 – Dec. 2013

Studied visual binary stars using speckle interferometry under Dr. Kent Clark at the University of South Alabama. Russell Genet of California Polytechnic State Institute led the study. Publication: Genet, R., [et al.]. (2015). *Kitt Peak Speckle Interferometry of Close Visual Binary Stars*. *Journal of Double Star Observations* 11(1s).

Spectroscopy Research

Aug. 2011 – Dec. 2012

Used gamma-ray spectroscopy to date soil core samples by measuring the abundance of Cs_{137} , Pb_{210} , and other radioactive isotopes under Dr. Justin Sanders at the University of South Alabama.